Subj: **RE: Quantum Standing Waves**
Date: 2/23/2003 6:57:16 PM Eastern Standard Time
From: [redacted]
[redacted]net
To: [redacted]
CC: [redacted]

P.S.

Steve,

In writing in some haste earlier on my way out the door, I hope it came through tha[redacted] insight is remarkable, because it is insight into some very advanced physics not usually drawn together. It is also in areas that are still recognized as some of the toughest problems in physics. So Bugh's work shows him to be an electrical engineer with remarkable native intuition and insight -- far more than one will see in the usual electrical engineer. I'm sure you're aware that most EEs simply never bother to even consider such things or to question them. There are two ways of thinking: (1) IN the model, and (2) OUTSIDE the model, where the "model" is the proverbial "box" everyone uses in the buzz words, "thinking outside the box". Too many of our engineers and scientists simply accept the model (box) taught to them as being perfect, and then never question things deeply again. In Bugh's case, he is not in that category (1), but is definitely in category (2).

E.g., NOBODY knows what really creates the arrow of time. Like many others, I have proposed a tentative approach, but that does not mean it's proven or even that it is necessarily correct. So Bugh's insight has to be very keen, to lead him directly into these forefront areas. And he is quite correct that ANY progress in these areas is very important indeed.

I've been working, e.g., on the arrow of time for some years, and it's a very strange bear. Seems to involve several things that got "left out" or "dropped along the wayside" accidentally. I put together a little of that in the Energy from the Vacuum book, but much still needs doing.

Even the spins are still an area with controversy. When you come right down to it, the nature of spin is still somewhat of a mystery. Most physicists just apply the equations, and let the concept of spin take care of itself. But to try to put into words what it really is, is very difficult and probably doesn't even come out correct. Spin is also remarkable because, if we try to view it as real "spin" and just ordinary angular momentum like a physical rotation in 3-space, the charge "spins" 720 degrees -- not 360! -- before returning to where it started! So right away spin is telling one that one's native 3-dimensional reasoning cannot get at 720 degrees of rotation between beginning state and return to it. You spin all you can in 3-dimensions, and then you continue the spin "outside" three dimensions by the same amount, and there you are back to the beginning in 3-space again! So it is not just 3-spatial rotation, but maybe something like "4-spatial" rotation???? Something like spinning 360 degrees while also "turning upside down", and then spinning another 360 degrees while one "turns upside down again, to wind up back where one started." No one really has a good handle on it, except just with equations.

I was struck that he BEGAN with considering the nature of time, probably quite a few years ago. That's one of the toughest problems of all. The very essence of thermodynamics, e.g., is up for grabs if one can manipulate or affect that arrow of time. "Fluctuations", e.g., are a mental crutch to cover the situation where the arrow of time momentarily "fluctuates" or reverses, and instead producing positive entropy we momentarily have negative entropy produced. So then we are faced with, "Why did it "fluctuate" at all, if entropy is a law of nature????" Some law! It's true except when it isn't, or decides not to be true for a bit. And so on. With the work last year by D. J. Evans et al., it isn't true in some little volumes in liquids for up to 2 seconds in duration and up to a cubic micron in size. In water, e.g., a cubic micron has some 30 billion water molecules in it. So a cubic micron of fluid with 30 billion ions and molecules, where the reaction(s) are momentarily (for up to two seconds!) running backwards is NOT a trivial or small effect! It has tremendous implications for chemistry, e.g., as pointed out by Evans.

Also, to go into spins and spin flipping on his own, shows a native intuition and an instinct that hones in where the action really is.

These are areas that are forefront, and in which some real breakthroughs are occurring and many more should occur in the future.

Just wanted to get that in, to let you know how I viewed the paper and the keen insight of the author. Please encourage Mr. Bugh to continue thinking! Really innovative thinking is a product in remarkably short supply these days!

Best wishes,
[redacted]

-----Original Message-----
From: [redacted] [mailto[redacted]]
Sent: Sunday, February 23, 2003 11:02 AM
To: [redacted]
Cc: [redacted]; s[redacted]
Subject: Fwd: Quantum Standing Waves

[redacted]
What do you think of these concepts. Have you heard of this gentleman? Best always, Steven

---------- Headers ----------
[redacted]

Subj: **RE: Quantum Standing Waves**
Date: 2/23/2003 7:20:02 PM Eastern Standard Time
From: [redacted]
Reply-to: [redacted]t
To: [redacted]

PPS.

One last comment: Actually, the setup to evoke the spin flips and exchange forces where and when in the rotation cycle one wishes, and in the direction one wishes, is a strange sort of legitimate "Maxwell's Demon". Such demons can in fact be made, and that has now been proven experimentally.

E.g., if one places a symmetrical loop of very tiny nanowire conductor into Brownian motion of ions (charges), one gets spurious voltages and currents in the loop that average out to zero. No power there to be had. But if one "flattens" one side of the loop a little bit, voila! The symmetrical side acquires a DC voltage and current, and it "powers" the asymmetric flattened side as an electrical load. So it TAKES controlled energy (consumes positive entropy to make negative entropy) from the seething random Brownian motion, then changes its form to perform work — some 10exp(-8) watts — in the "load" — with the change in form of the energy being a dissipation (positive entropy) back into the seething Browning motion "sea". Russian scientists have quite a few papers and experiments now, proving that one conclusively.

See, e.g., Alexey Nikulov, "Quantum Power Source", which I found on http://www.chronos.msu.ru/EREPORTS/nickulov.htm. It contains the detailed explanation, etc. and also additional references in the published literature. This is really good scientific work. It appears that some 10exp8 of these could be placed on a surface about 1 cm square, to provide one watt of power for free. By multilayering the squares, one could build up a little power supply of, say, 100 watts. It could sit there and power a 100 watt load continuously, with no problem. The work was supported by the Russian Academy of Sciences, and is part of their program on "Low Dimensional Quantum Structures".

Cheers,
[redacted]

—–Original Message—–
From: [redacted] [mail[redacted]
Sent: Sunday, February 23, 2003 5:47 PM
To: [redacted]t; [redacted]
Cc: [redacted] Waves

Thank You [redacted] Great explanatiion; appreciate the input. Best to you and your
wife, Steven

———————— Headers ————————
Return-Path: [redacted]>